INCH-POUND

MIL-M-48186B(AR)
14 November 1990

SUPERSEDING
MIL-M-48186A(AR)
12 August 1977

MILITARY SPECIFICATION

MUNITION FOR PROJECTILE, 155MM, HE, M692/M731, LOADING AND ASSEMBLING OF

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading and assembling of the munition used in the Projectile, 155MM, HE, M692/M731.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-14970	- Explosive Composition A5
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-M-48245	- Molding Compound, Housing and Plug

STANDARDS

FEDERAL

FED-STD-595	- Colors

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1320

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9298400 - Mine, Anti-personnel, (Type I or Type 2)
9298699 - Delay Detonator Functional Test Configuration
9298700 - Firing Train Test Mine, HE

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Non-government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specifed, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D-695 - Plastics

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with the applicable drawings and specifications.

3.2 Mine. The mine shall comply with all requirements specified on drawing (dwg. 9298400), all associated drawings, and with all requirements specified in applicable specifications.

3.3 Documentation of controls for use of adhesives and sealants. The contractor shall develop a documented operational procedure for utilization of adhesives and sealants. This procedure shall provide for control of the materials and processes to be used; including controls for the following: shelf life, handling, mixing, curing and pot life. Necessary cleaning agents, safety precautions, ventilation, and any other information considered necessary for the proper use of the adhesive(s) or sealant(s) shall also be included. The procedure shall be updated whenever a change is made to the process such that product quality may be affected. Submission of this procedure shall be in accordance with Data Item Description DI-T-5372 (Tailored).

3.4 Static functioning test of kill mechanism and overlay assembly. The kill mechanism and overlay assembly shall function satisfactorily as evidenced by compliance with all the following:

a. The test housing containing the kill mechanism and overlay assembly (K.M. & O.) shall be broken and entirely separated from the kill mechanism assembly.

b. The kill mechanism assembly shall be projected upward from the test housing and shall explode its main charge high order, i.e., no kill mechanism fragment shall be larger than 0.25 inches in any direction.

c. The average height at which the main charge explodes shall be not less than 0.5 feet nor more than 8 feet above the surface upon which the housing was resting.

d. Average burst height shall be between 2.5 to 8.0 feet

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Only those lots of kill mechanism and overlay assemblies which have met the acceptance criteria in Table II for the static functioning test shall be assembled into the housing, timing and fuzing.

3.5 Qualification test of delay detonator and propellant nut assembly serial lots. The delay detonator serial lot and propellant nut assembly serial lot to be used in the kill mechanism and overlay assembly lot shall perform satisfactorily when tested together as evidenced by compliance with the qualification test criteria specified in 4.5.5.

3.6 Documentation of plan for safety in handling and processing of munitions. Prior to any loading, the contractor shall document a detailed written plan of the procedures to be employed during handling of the munitions. This documentation shall describe as a minimum, procedures to be employed to assure that:

a. The shorting bar is not deformed or broken.

b. The battery ball is not depressed.

c. The liquid propellant does not leak externally from the overlays.

d. The safing and arming (S&A) barrier assemblies are present and properly positioned in each housing prior to loading.

e. The Type 1 and Type 2 housing, timing and fuzing (HTF) and munitions remain segregated at all times.

f. The temperature of the housing, timing and fuzing or munitions does not exceed 165°F at any time during storage or processing.

g. HE and inert munitions remain properly identified and segregated at all times.

Any changes to the approved plan shall be submitted for approval, prior to implementation. Submission of this plan shall be in accordance with Data Item Description DI-T-5372 (Tailored).

3.7 Bulk density, moisture content and consolidation pressure of explosive composition A5. The contractor shall provide a means of automatically and continually monitoring and recording the consolidation pressures on each machine used in loading the upper and lower shell assemblies and shall adjust charge weights based on

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the recorded pressures, such that the consolidation density remains constant with volume changes in the lots of shell metal parts. The bulk density of the explosive complosition A5 loaded into the upper and lower shell assemblies shall be not less than 0.83 nor more than 1.00 grams/cc as determined by the "Bechtel Method". The moisture content of the explosive composition A5 shall not exceed 0.10 percent at the time of loading.

3.8 Documentation of plan for molding and cure process controls. The contractor shall prepare a written plan to be implemented for control of the process for molding the kill mechanism and overlay assembly and the plug into the housing, timing, and fuzing. The plan shall briefly describe the entire process and shall fully document the controls to be implemented for maintaining the process. The controls shall include, as a minimum, all the checkpoints referenced in 6.4. Submission of this plan shall be in accordance with Data Item Description DI-T-5372 (Tailored).

Approval of the plan shall be limited to the controls to be employed and not the overall potting process, nor shall such approval relieve the contractor, in any way, of the responsibility for producing acceptable product.

3.9 Molding compound for casting or assembling plugs into housings. The molding compound used for casting into the housing shall comply with MIL-M-48245. The resin/hardener weight ratio shall be 2.0/1.0 $\pm$ 5% (1.8/1 to 2.2/1) and the ultimate compressive strength shall be 29,000 psi. minimum.

3.10 Temperature control during molding and cure of munition. The housing, timing and fuzing, or munition shall not be subjected to temperatures exceeding 165°F at any time.

3.11 Color of arc face. The color of the arc face on the completed munition shall be within the range of colors bounded by color numbers 34079 and 34102 inclusive of FED-STD-595.

3.12 Data requirements. The contractor shall generate data in accordance with the requirements of the data item descriptions cited in 3.3, 3.6, 3.8, 4.4.2, 4.5.14.6, 6.8.

3.13 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.14 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, discoloration and other defects which would make the item unsuitable for its intended use.

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4 QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First Article Inspection.

b. Quality Conformance Inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Propellant Nut Assembly	9298433	50
Overlay Assembly	9298412	25
Kill Mechanism and Overlay Assembly, HE	9298405	25
Mine, Anti-personnel, (Type 1 or Type 2)	9298400	25
Firing Train Test Mine, HE	9298700	85

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

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4.3.3 Rejection. See MIL-A-48078. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon failure of any assembly, component or test specified in the sample to comply with any of the stated requirements.

NOTE: The Government reserves the right to terminate its inspection with any failure of an assembly, component or test specimen in the sample causes rejection per the accept-reject criteria for that examination or test.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186B (AR)

PARAGRAPH	TITLE: Munition and components for Proj., 155MM, HE, M692/M731	SHEET 1 OF 1	DRAWING NUMBER: See below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Propellant Nut Assembly (Dwg. 9298433)				
	Examination for Defects	50		3.2	4.4.2.6
	Qualification	50		3.2	4.5.5
	Overlay Assembly (Dwg. 9298412-1)				
	Examination for defects	25		3.2	4.4.2.19
	Leakage Test	25		3.2	4.5.9
	Kill Mechanism and Overlay Assembly HE (Dwg. 9298405)				
	Examination for Defects	25		3.2	4.4.2.21
	Leakage Test	25		3.2	4.5.10
	Mine, Anti-personnel, Type 1 and Type 2 (Dwg. 9298400)				
	Examination for Defects	25		3.2	4.4.2.25
	Firing Train Test Mine, HE				
	Static Functioning Test	85 See Note		3.4	Table II and 4.5.14

NOTES:
Five(5) mines are intended for use as replacement in the event of arming failures.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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TABLE II
Arena Testing

Orientation (Face Resting on Pad)

Face #	Description	First Article -25°F	First Article Ambient	First Article +125°F	Production -25°F	Production Ambient	Production +125°
5	Arc Face (large radius)	6	20	6	4	12	4
4	Slant Face with blue & yellow leads (PA)	3	10	3	2	6	2
3	Slant Face with red & yellow leads (Det)	3	10	3	2	6	2
2	Triangular face without safety screw	4	8	4	2	6	2
		16	48	16	10	30	10
			80 Total			50 Total	

Rejection

First Article - (Sample size of 80- test wedges)
The sample quantity shall be rejected if any one of the following occurs:

a. Five or more test wedges fail to comply with any of the requirements specified in 3.4. (80-4-5)*.

b. The average burst height of all units except those tested on face 1 fails to comply with the requirements specified in 3.4.

Production - (Sample size of 50 test wedges)
The lot shall be rejected if any one of the following occurs:

a. Four or more test wedges fail to comply with any of the requirements specified in 3.4. (50-3-4)*

b. The average burst height of all units except those tested on face 1 to comply with the requirement specified in 3.4.

*NOTE. No more than two (2) failures shall occur at any test temperature or in any orientation.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Inspection lot of mines shall contain:

a. Housing, timing and fuzing of one lot serial number from one manufacturer and of one type only.

b. Shell tube assemblies (8865570) of one lot interfix number from one manufacturer.

c. Delay detonators (9298429) of one lot interfix number from one manufacturer.

d. Overlay assemblies from one lot interfix number from one manufacturer.

e. Detonating cord assemblies (9298406) of one lot interfix number from one manufacturer.

f. Explosive composition A5 from one batch or blending of batches which results in homogneity of all composition used in loading the upper and lower shell assemblies.

g. Propellant (9298437) and WC-140 from one batch or blending of batches which results in homogeneity of all propellant used in loading the propellant nut assembly.

h. Propellant nut assemblies from one lot.

i. Liquid propellant from one batch or blending of batches which results in homogeneity of all composition used in loading the overlay assemblies.

j. Lower shells (8865569) on one lot interfix number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Nut, Propellant		9298434
			NEXT HIGHER ASSEMBLY 9298433

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hole missing or blocked		0.015%	3.2	Visual
102	Diameter of hole		0.40%	3.2	Gage
103	Location of hole from center of nut		0.40%	3.2	Gage
104	Width of groove		0.40%	3.2	Gage
105	Depth of groove		0.40%	3.2	Gage
106	Flange thickness		0.40%	3.2	Gage
107	Distance from shoulder face to bottom of small diameter cavity		0.40%	3.2	Gage
108	Distance from shoulder face to bottom of large diameter cavity		0.40%	3.2	Gage
109	Diameter of small cavity		0.40%	3.2	Gage
110	Diameter of large cavity		0.40%	3.2	Gage
111	Major diameter of thread		0.40%	3.2	Gage
112	Pitch diameter of thread		0.40%	3.2	Gage
Minor					
201	Diameter of flange		0.65%	3.2	Gage
202	Poor workmanship		0.65%	3.14	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Plate, Oriface		9298435
			NEXT HIGHER ASSEMBLY 9298433

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hole blocked or missing		100%	3.2	Visual
102	Diameter of hole		0.40%	3.2	Gage
103	Thickness		0.40%	3.2	Gage
Minor					
201	Diameter		0.65%	3.2	Gage
202	Poor workmanship		0.65%	3.14	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Propellant (prior to loading into propellant nut)	1 OF 1	9298437
			NEXT HIGHER ASSEMBLY 9298433

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Propellant mixture incorrect			3.2	Certification
102	Moisture content of propellant			3.2	Note 1, 2 4.5.3
Minor					
201	Poor workmanship		0.65%	3.14	Visual

NOTES:
1. Certification shall include the actual percentage by weight of each constituent immediately prior to blending.
2. The propellant mix shall be lotted and a Propellant Description Sheet (ARMCOM-214-R) shall be prepared for each lot. The sheet shall be available for review by the Government.

REMARKS:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Propellant nut (prior to loading into propellant)		9298434
			NEXT HIGHER ASSEMBLY 9298433

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thread damaged		0.40%	3.2	Visual
102	Foreign matter in any cavity or hole		0.40%	3.2	Visual
103	Groove missing		0.40%	3.2	Visual
104	Seal at base of propellant cavity missing		0.40%	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.14	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Propellant nut assembly (prior to loading into propellant)		9298433
			NEXT HIGHER ASSEMBLY 9298433

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Weight of propellant			3.2	Note A
Minor 201	Poor workmanship		0.65%	3.14	Visual
NOTE A:	The Propellant Nut Assembly machine will be qualified at the beginning of each production interval by processing a minimum of two (2) acceptable weight samples prior to use in production. The machine will be requalified each time it is removed from the production process for maintenance or adjustment. A minimum of two (2) samples shall be processed each hour of production or fraction thereof. Each hour's production or fraction thereof shall be accepted provided all samples processed during that period meet weight requirements. If a sample fails to meet weight requirements, all loaded propellant nut assemblies processed since the last acceptable check shall be rejected. The assembly machine shall be stopped for corrective action and requalified prior to release for production.				

NOTES:

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Propellant Nut Assembly	1 OF 1	9298433
			NEXT HIGHER ASSEMBLY: 9298417

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
1	Propellant present on threads		100%	3.2	Visual
2	Propellant present on other surface outside of the propellant cavity		100%	3.2	Visual
Major					
101	Seal under orifice plate missing		0.40%	3.2	Visual
102	Stake of orifice plate missing (4 places)		0.40%	3.2	Visual
103	Foreign matter or burr in hole of orifice plate		0.40%	3.2	Visual
104	Force test of stake			3.2	4.5.4
105	Qualification test of propellant nut assembly and delay detonator	50 1/		3.5	4.5.5/9298699
Minor					
201	Poor workmanship		0.65%	3.14	Visual

NOTES:
1. The samples shall be selected from the machine at the beginning of production, and equal intervals during production of the lot and from the last portion of production of the lot.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Explosive Composition A5, prior to loading	1 OF 1	9298357
			NEXT HIGHER ASSEMBLY 9298357

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bulk density		---	3.7	4.5.2.1
102	Moisture content		---	3.7	4.5.1
Minor					
201	Poor workmanship		1.0%	3.14	Visual

NOTES:

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Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Upper Shell Assembly		9298357
			NEXT HIGHER ASSEMBLY 9298417

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Presence of sleeve, outer, positioned over tube assembly		100%	3.2	Visual
Major					
101	Sleeve height		0.40%	3.2	Gage
102	Depth to charge		0.40%	3.2	Gage
103	Outside diameter, max.		0.40%	3.2	Gage
104	Diameter of detonator tube sleeve, min		0.40%	3.2	Gage
105	Inside diameter of detonator tube		0.40%	3.2	Gage
106	Charge chipped, cracked or pitted		0.40%	3.2	Visual
107	Density of charge			3.2	4.5.2
108	Spring grip fastener missing		0.40%	3.2	Visual
109	Consolidation pressure			3.7	Certification
110	Presence of exuded adhesive (360°) on top of sleeve		0.40% (1)	3.2	Visual
111	Outer sleeve secure		Note 1	3.2	Manual
Minor					
201	Poor workmanship		1.0%	3.14	Visual

NOTES: (1) In accordance with MIL-STD-1235.

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MIL-M-48186B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Lower Shell Assembly	1 OF 1	9298358
			NEXT HIGHER ASSEMBLY 9298417

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Depth to charge		0.40%	3.2	Gage
102	Outside diameter, max		0.40%	3.2	Gage
103	Diameter of hole in shell		0.40%	3.2	Gage
104	Diameter of hole in charge		0.40%	3.2	Gage
105	Charge chipped, cracked or pitted		0.40%	3.2	Visual
106	Density of charge			3.2	4.5.2
107	Consolidation pressure			3.7	Certification
108	Comp. A5 shoulder, 360° not present		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.14	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Delay Detonator	1 OF 1	9298429
			NEXT HIGHER ASSEMBLY 9298417

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Qualification test	50		3.5	4.5.5/9298699
102	Delay composition not below flush		1.0%	3.1.4	Visual
103	Delay composition cracked or chipped		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.14	Visual

NOTES:

AMSMC ( ) 1570, 1 Feb 85 Replac DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used. (

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Kill Mechanism Assembly				9298417
					NEXT HIGHER ASSEMBLY 9298405
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter, max		0.40%	3.2	Gage
102	Total height, max		0.40%	3.2	Gage
103	Foreign matter on exterior surface (explosive, excessive sealant, etc)		0.40%	3.2	Visual
104	Sealant missing on any portion of joint between upper and lower shell assemblies		0.40%	3.2	Visual/4.5.6
105	Torque of propellant nut assembly		0.40%	3.2	4.5.7
Minor					
201	Poor workmanship		0.65%	3.14	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Seal, Upper Shell (prior to assembly to kill mechanism)	1 OF 1	9298418
			NEXT HIGHER ASSEMBLY 9298417

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pad missing		0.40%	3.2	Visual
102	Hole in pad missing		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.14	Visual

NOTES:

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MIL-M-48186B(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.13	Overlay, Inner	1	1	9298414
				NEXT HIGHER ASSEMBLY: 9298412-1

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material thickness	S4	0.40%	3.2	Gage
102	Profile of surface between flange and hole	S4	0.40%	3.2	Gage
103	Presence of cracks		0.40%	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.14	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Overlay, Outer	1 OF 1	9298413
			NEXT HIGHER ASSEMBLY: 9298412-1

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Profile of surface between flange and hole	S4	0.40%	3.2	Gage
102	Depth to bottom of groove (2 places)	S4	0.40%	3.2	Gage
103	Material thickness	S4	0.40%	3.2	Gage
104	Presence of cracks		0.40%	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.14	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.15	Spacer, Overlay	1 OF 1	9298415
			NEXT HIGHER ASSEMBLY: 9298412-1

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of hole		0.40%	3.2	Gage
102	Largest outside diameter		0.40%	3.2	Gage
103	True position of largest outside diameter with hole		0.40%	3.2	Gage
104	Thickness over circular and rectangular protrusions (12 places)		0.40%	3.2	Gage
105	Thickness adjacent to protrusions		0.40%	3.2	Gage
Minor					
201	Presence of flash		0.65%	3.14	Visual
202	Poor workmanship		0.65%	3.14	Visual

REMARKS:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Overlay, Welded (Prior to welding)	1 OF 1	9298421
			NEXT HIGHER ASSEMBLY: 9298412-1

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor 201	Poor workmanship		0.65%	3.14	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.17	Overlay Welded		9298421
			NEXT HIGHER ASSEMBLY 9298412

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Maximum outside diameter		0.40%	3.2	Gage
102	Minimum inside diameter of equator		0.40%	3.2	Gage
Minor					
201	Minimum depth of inside diameter		0.65%	3.2	Gage
202	Poor workmanship		0.65%	3.14	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	Overlay Assembly (Prior to trim of polar hole)	1 OF 1	9298412-1
			NEXT HIGHER ASSEMBLY 9298405

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Liquid propellant on exterior of overlay (indicating spillage during fill)			100%	Visual
102	Quantity of liquid propellant incorrect		0.40%	3.2	Balance Note 1
103	Liquid propellant formulation incorrect				Note 2, 3

NOTES:

1. The quantity of liquid propellant shall be determined by weight difference of the overlay before and after loading of the propellant.
2. The liquid propellant shall be lotted and a Propellant Description Sheet (ARMCOM-214-R) shall be prepared for each lot. The propellant shall comply with MIL-P-48333.
3. Alternate: Machine control. Liquid propellant dispensing machines will be qualified at beginning of each production interval, each production hour, and each time the machine is removed from the production for maintenance or adjustment. A minimum of two (2) samples will be dispensed and weighed for qualification. Each hour's production or fraction thereof will be accepted or rejected based upon the qualification samples.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.19	Overlay Assembly		9298412-1
			NEXT HIGHER ASSEMBLY 9298405

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Leakage test		100%	3.2	4.5.9
102	Height of overlay		0.40%	3.2	Gage
103	Diameter of hole		0.40%	3.2	Gage
104	True position of hole with inside diameter at equator		0.40%	3.2	Gage
105	Assembly damaged		0.40%	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.14	Visual

REMARKS:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.20	Detonating Cord Assembly (prior to assembly to kill mechanism and overlay assembly)	1 OF 1	9298409
			NEXT HIGHER ASSEMBLY: 9298405

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cord sheath cracked or broken		0.40%	3.2	Visual
102	Booster assembly damaged		0.40%	3.2	Visual
103	Kink or sharp bend in cord		0.40%	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.14	Visual

NOTES:

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.21	Kill Mechanism and Overlay Assembly (prior to assembly into housing)	1 OF 2	9298405
			NEXT HIGHER ASSEMBLY 9298400

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Evidence of leakage of overlay assembly		100%	3.2	4.5.10
102	Height of detonating cord above propellant nut		0.40%	3.2	Gage
103	Outside diameter, max (at equator)		0.40%	3.2	Gage
104	Distance across poles of overlay assemblies, max.		0.40%	3.2	Gage
105	Grooves in overlays not aligned with groove in propellant nut		0.40%	3.2	Visual
106	Detonating cord twisted, kinked, cracked or broken		0.40%	3.2	Visual
107	Booster improperly positioned for assembly to S&A		0.40%	3.2	Visual
108	Detonating cord not seated in grooves of overlays and propellant nut		0.40%	3.2	Visual
109	Sealant inadequate or missing from detonating cord assembly and propellant nut interface		0.40%	3.2	Visual
110	Upper seal missing		0.40%	3.2	Visual
111	Static functioning test	50		3.4	4.5.14 and Table 2

NOTES:

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MIL-M-48186B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.21	Kill Mechanism and Overlay Assembly (prior to assembly into housing)	2 OF 2	9298405
			NEXT HIGHER ASSEMBLY 9298400

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Tape around equator of overlay assemblies missing, loose or not positioned correctly		0.65%	3.2	Visual
202	Poor workmanship		0.65%	3.14	Visual

NOTES:
Data Item Description DI-R-1721 applies to all Critical, Special, and 100% Major characteristics. Submission of the report shall be on a weekly frequency.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.22	Housing, Timing and Fuzing (prior to assembly of kill mechanism and overlay assemblies	1 OF 1	9298450
			NEXT HIGHER ASSEMBLY 9298400

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Barrier assembly not present or not properly positioned in S&A		100% 1/	3.2	X-ray
2	Shorting bar broken of damaged		100%	3.2	Visual
Major					
101	Detonator shield missing loose		100%	3.2	Visual
102	Battery ball depressed below normal position		100%	3.2	Gage
103	Bare metal exposed in kill mechanism cavity		100%	3.2	Visual
104	Foreign material in kill mechanism cavity		0.40%	3.2	Visual
105	Largest diameter in kill mechanism cavity not prepared for bonding		0.40%	3.2	Visual
106	Any external component, including sensor locating plugs, missing or damaged		0.40%	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.14	Visual

1/ X-ray inspection is required if not previously performed by HTF assembly manufacturer.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298404
4.4.2.23	Keeper				NEXT HIGHER ASSEMBLY 9298400
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Height (.22 - .01) maximum		0.40%	3.2	Gage
102	Width (.190 - .005) maximum		0.40%	3.2	Gage
Minor					
201	Presence of flash		0.65%	3.14	Visual
202	Poor workmanship		0.65%	3.14	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.24	Mine (prior to cast filling)	1 OF 1	9298400
			NEXT HIGHER ASSEMBLY: 9298400

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Distance from apex to top of kill mechanism and overlay assembly, max		0.40%	3.2	Gage
102	Booster of detonator cord assembly protruding beyond keeper in S&A		0.40%	3.2	Visual
103	Keeper not seated on booster		0.40%	3.2	Visual
104	Bedding compound improper		0.40%	3.2	4.5.11
105	Retaining clips missing or incorrectly positioned		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.14	Visual

NOTES:

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MIL-M-48186B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.25	Mine, Antipersonnel	1 OF 2	9298400
			NEXT HIGHER ASSEMBLY: N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Shorting bar broken or damaged		100%	3.2	Visual
Major					
101	Evidence of liquid propellant leakage		100%	3.2	Visual
102	Battery ball below minimum		0.40%	3.2	Gage
103	Shorting bar height below minimum		0.40%	3.2	Gage
104	Total height		0.40%	3.2	Gage
105	Arc face radius		0.40%	3.2	Gage
106	Distance from arc face to apex gaging radius		0.40%	3.2	Gage
107	Perpendicularity of arc face to triangular faces		0.40%	3.2	Gage
108	Surface finish on arc face		0.40%	3.2	Finish blocks
109	Any sensor deployed		0.40%	3.2	Visual
110	Voids in plug bond area		0.40%	3.2	Visual
111	Any sensor restricted (cap not removed or molding compound flash over sensor cutout)		0.40%	3.2	Visual
112	Temperature control during cure		0.40%	3.10	4.5.13

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.25	Mine, Antipersonnel	2 OF 2	9298400
			NEXT HIGHER ASSEMBLY: N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major	Cont'd				
113	Resin/hardener weight ratio of molding compound		0.40%	3.2	4.5.12
114	Compressive strength of molding compound		0.40%	3.2	4.5.12
115	Molding compound shelf life expired		0.40%	3.2	Visual
116	Arc face color incorrect		0.40%	3.11	Visual
117	Sealing compound not at proper level		0.40%	3.2	Visual
Minor					
201	Test pads exposed		0.65%	3.2	Visual
202	Poor workmanship		0.65%	3.14	Visual

NOTES:
Data Item Description DI-R-1721 applies to Critical and 100% Major Characteristics. Submission of the report shall be on a weekly frequency.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.26	Sleeve, Outer		9298356
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Sleeve, length		0.40%	3.2	Gage
102	Sleeve, inner diameter		0.40%	3.2	Gage
103	Sleeve, outer diameter		0.40%	3.2	Gage
104	Sleeve, concentricity		0.40%	3.2	Gage
105	Sleeve, chamfer		0.40%	3.2	Gage
106	Sleeve, hardness	---		3.2	Certification
107	Sleeve, finish		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.14	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.27	Explosive Composition A5, prior to loading	1 OF 1	9298358
			NEXT HIGHER ASSEMBLY: 9298358

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bulk density			3.7	4.5.2.1
102	Moisture content			3.7	4.5.1
Minor					
201	Poor workmanship		1.0%	3.14	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078.

4.5 Methods of inspection.

4.5.1 Moisture content of explosive composition A5 (for loading upper and lower shell assemblies). The contractor shall provide adequate controls to insure compliance with the applicable requirement. The moisture content of the explosive composition shall be determined at the loading station at the time of loading. This determination shall be made at least once each eight (8) hours of production. Sampling of the explosive composition shall be in accordance with MIL-E-14970. In addition, if the moisture content fails to comply with the requirement, and loading has not begun, the explosive composition shall be rejected until corrective action has been taken as verified by repeating the moisture content determination. If loading has begun, all assemblies loaded with the non-conforming explosive composition shall be rejected. The moisture content shall be determined in accordance with MIL-E-14970.

4.5.2 Density of explosive composition A5.

4.5.2.1 Bulk density of explosive composition. The contractor shall provide adequate controls to insure compliance with the applicable requirements. The bulk density of each batch of the explosive composition shall be determined at the loading station at the time of loading. This determination shall be made on each bucket on each loading machine immediately priro to its use. Sampling shall be in accordance with MIL-E-14970. In addition, if the bulk density fails to comply with the requirement specified on the applicable drawing and loading has not begun, the composition lot shall be rejected until corrective action has been taken as verified by repeating the bulk density determination. If loading has begun, assemblies with the non-conforming composition shall be rejected. The following procedure shall be used to determine the bulk density of the explosive composition.

a. Use 100cc capacity cup (metal cup recommended), polished inside and outside to eight (8) micro-inch finish, and a tare weight equal to the weight of the empty cup. The dimensions of the 100cc cup shall be:

I.D. = 4.0cm Inside depth = 7.958 cm
Inside bottom radium = 0.076 mm, max.
Wall thickness = 0.2 cm (approx.)

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b. Place both 100cc cup and the tare weight on analytic balance scale, or equal, one on each pan and check that cup and tare weight balance.

c. Set cup on level surface and fill with composition to be bulk-density tested. To fill, use a scoop, approximately tablespoon size, or a tablespoon, and fill the cup above the rim. Do not tamp or shake at any time. Level off excess composition with a straight edge, keeping side of straight edge of vertical to rim of cup.

d. Place filled cup on balance pan. Add gram weights to other pan with tare weight until balance is reached. Number of grams added will give the bulk density in percent.

4.5.2.2 Consolidation density of charge in upper and lower shell assemblies. A minimum of two (2) samples each, representative of both the shell tube assembly and the lower shell being processed, shall be loaded and the densities determined to qualify each press prior to its use in the loading process in production. In addition, a sample of two (2) assemblies shall be randomly selected from the last portion of each one (1) hour's production from each press and the density determined as a control of each press during production.

If any defective assemblies are found when inspection is performed for control of a press, the press producing the defective assembly shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All presses removed from production because of some fault, may, after corrective action, be returned to production, providing they pass the qualification test above. The following procedure shall be used to determine the density of the explosive composition.

When weighing in water, use distilled water containing a suitable wetting agent, e.g. "Aerosol" (1cc per 250cc distilled water).

a. Prior to loading, weigh the lower shell or upper shell as applicable.

b. Prior to assembly, weigh the same component in water (grams).

c. Thoroughly dry the component from b. (above) and assemble with explosive composition using normal production methods.

d. Weigh the loaded component in air (grams).

e. Weigh the same loaded component in water (grams).

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f. Calculate the weight of the explosive composition in air (d) - (a) (grams).

g. Calculate the weight of the explosive composition in water (e) - (b) (grams).

h. Calculate the weight (and volume) of water displaced by the explosive composition (f) - (g) (grams x 1cc per gram).

i. Calculate the density of the loaded explosive composition (f) / (h) (grams per cc).

4.5.3 Moisture content of propellant blend for propellant nut assembly. The contractor shall provide adequate controls to insure compliance with the applicable requirements. The moisture content of the propellant shall be determined at the loading station at the time of loading the propellant nut assemblies. This determination shall be made at least once for each lot of propellant nut assemblies. Sampling and testing of the propellant shall be in accordance with Method 101.1.1 of MIL-STD-286. If the moisture content fails to comply with the requirement specified on the applicable drawing and loading has not begun, the propellant shall not be used until corrective action has been taken as verified by repeating the moisture content determination. If loading has begun, assemblies loaded with the non-conforming propellant shall be rejected. Method T103.5 of MIL-STD-286 may also be used for this test.

4.5.4 Security of orifice plate in propellant nut assembly. At the start of production each day, each machine to be used for staking the orifice plate in the propellant nut shall be used to assemble two (2) inert (empty) propellant nut assemblies to qualify each machine for production. In addition, two (2) inert (empty) assemblies per machine shall be assembled after each hour's production or portion thereof, as a control of the equipment during production. If any defective assembly is found during qualification of the equipment, the equipment producing the defective assembly shall not be used in production. If any defective assembly is found when inspection is performed for control of the equipment, the equipment which produced the defective assembly shall be removed from production. In addtion, that portion of propellant nut assemblies produced by the equipment since the last check shall be rejected and returned to the contractor for corrective action.

All equipment removed from production because of some fault, may, after corrective action, be returned to production providing it passes the qualification test above.

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The inert (empty) assemblies shall be tested by drilling a 0.110 inch diameter hole (#35 drill) through only the head of the propellant nut and aligned with the powder cavity hole. The specified force shall be applied through a .109 - .005 inch diameter rod to the adhesive side of the orifice plate. If the orifice plate lifts from the propellant nut at any of the stakes after the specified force has been applied, the assembly shall be classed defective.

4.5.5 Qualification test of delay detonator serial lot and propellant nut assembly serial lot.

4.5.5.1 Test procedure. Prior to assembly of the Kill Mechanism and Overlay Assembly lot, delay detonators and propellant nut assemblies from the respective serial lots to be used in the kill mechanism and overlay assembly lot shall be subjected to this test. The test shall be performed using the method specified on Dwg. 9298699. Twenty (20) of the samples shall be tested at -25 ± 5°F, twenty (2) of the samples shall be tested at 125 ± 5°F. The propellant nut assemblies and delay detonators shall be conditioned at the specified test temperatures for 2 hours minimum prior to testing. All test data shall be recorded.

4.5.5.2 Scoring criteria. The combination of delay detonator serial lot and propellant nut assembly lot shall be rejected for use in loading if any of the following occur:

a. If the combined total of two (2) or more detonators (1) fire in the out-of-line position.

b. If four (4) or more detonators fire prior to or later than the specified delay time.

4.5.5.3 Limits of number of tests permitted. The following criteria will be used to limit the number of times a delay detonator or propellant nut lot can be tested:

a. If the combination of propellant nut lot and delay detonator lot passes the test, according to scoring critera in 4.5.5.2 use the combination in the Kill Mechanism and Overlay test.

b. If the combination of propellant nut lot and delay detonator lot fails the test, marginally, as per 4.4.5.2, with 4 failures, select an additional 50 samples and retest. The accept/reject criteria will be in accordance with the double sampling plan (100-7-8) using the results of the first 50 tested with the results of the second 50. No additional testing of the same propellant nut lot and delay detonator lot combination shall be permitted.

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c. Upon rejection according to (b) above, each propellant nut lot and delay detonator lot may be tested with two additional lots of delay detonators and propellant nuts, respectively, according to (b) above.

d. The propellant nut lot or delay detonator lot which fails the combination with three delay detonator lots or propellant nut lots, respectively, shall be rejected.

4.5.6 Adequacy of sealant of kill mechanism assembly. At the start of each shift's operation, five(5) completed kill mechanism assemblies shall be selected from each piece of equipment for determination of sufficiency of sealant indicated by a bead not less than 1/32 of an inch wide at contact of ball halves, 360° around, or as indicated upon disassembly by sealant completely covering the surface of the component in an area between the outer edge and three sixteenths (3/16) to five sixteenths (5/16) of an inch toward the center for qualifying the equipment. In addition, two(2) assemblies per machine shall be selected from each hour's production, or portion thereof, for determination of sufficient sealant as a control of the equipment. If any defective assemblies are found during qualification of the equipment, the equipment producing the defective assemblies shall not be used in production. If any defective assembly is found when inspection is performed for control of the equipment, the equipment which produced the defective assembly shall be removed from production. Further, that portion of production since the last equipment check, shall be returned to the contractor for corrective action.

All equipment removed from production because of some fault, may, after corrective action be returned to production, providing it passes the qualification test above.

4.5.7 Minimum assembly torque of propellant nut assembly. In place of the normal sampling procedures associated with the classification of defects and with the approval of the Contracting Officer, the following inspection procedure may be used: at the start of each day, and whenever a new piece of equipment is put into operation, five (5) kill mechanism assemblies shall be selected from each piece of equipment for determination of minimum assembly torque of the propellant nut assemblies for qualifying the equipment. In addition, five (5) kill mechanism assemblies per machine shall be selected from each hour's production or fraction thereof, for the determination of the minimum propellant nut assembly torque as a control of the equipment. If any defective assemblies are found during qualification of the equipment, the equipment producing the defective assembled shall not be used in production. If any

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defective mechanism assembly is found when inspection is performed for control of the equipment, the equipment having produced the defective assembly shall be removed from production. Further, that portion of production since the last equipment check shall be returned to the contractor for corrective action. All equipment removed from production because of some fault, may, after corrective action, be returned to procuduction providing it passes the qualification test above.

4.5.8 Control of quantity of propellant in overlay assembly. The contractor shall provide necessary controls to assure that the quantity of propellant loaded into each overlay assembly is within the requirement specified on the applicable drawing. The method of control shall be submitted for approval as specified in 4.4.4.

4.5.9 Leakage test of overlay assembly. The assemblies shall then be cleaned in an acetone bath and placed "equator down" on filter paper and filter paper placed on top of overlay assembly "polar hole" in a tray. The assemblies shall be subject to a vacuum of 10mm mercury (0.193 psia) for five minutes minimum and visual observation made for propellant leakage as evidenced by orange staining on the filter paper. The assemblies shall then be cleaned in an acetone bath and placed "equator down" on filter paper and filter paper placed on top of the overlay assembly "polar hole" in a tray. The assemblies shall then be placed in an oven at 160 $\pm$ 5°F for forty (40) hours minimum, and visually examined for any evidence of propellant leakage as evidenced by orange staining on the filter paper. All assemblies which have any evidence of leakage shall be classed defective and removed from the lot.

4.5.10 Leakage test of kill mechanism and overlay assembly. The assemblies shall be tested by conditioning in a preheated oven at 160 + 5°F for 15 minutes minimum. The assemblies shall then be removed from the oven and immediately subjected to a vacuum of 10MM mercury absolute pressure (0.193 psia) for 5 minutes minimum. After removal from the vacuum chamber the assemblies shall be visually inspected for evidence of liquid propellant leakage especially around the pole and equator welds of the overlay assemblies. Assemblies which show any evidence of leakage including staining of the tape around the equator, shall be classed defective and removed from the lot.

4.5.11 Quality controls for sealing compound.

4.5.11.1 Sealing compound batch. A batch of sealing compound is herein defined as that quantity of material mixed at one time for use in assembling kill mechanism and overlay assemblies in housings prior to plug molding.

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4.5.11.2 Certification of properties. Certification of compliance with the requirement for gel time, shall be available for each batch of sealing compound. In addition, the data from the test shall be furnished to the Government upon request. If Part A is mixed and stored, the tests shall be run after storage and immediately prior to use.

4.5.11.3 Records of addition of catalyst (part B). After the compound has been mixed and the samples for gel time have been taken records shall be maintained to show the amount of catalyst added and any other additions or adjustments made to the compound at any time until the batch is consumed.

4.5.11.4 Records of shelf life. When the compound is mixed and stored prior to use, containers shall be marked to show the date the compound was mixed and a date before which the compound must be used. In addition all containers with ingredients for the sealing compound shall be marked with the appropriate shelf life dates. Pre-mixed "Kits" of ingredients shall also be marked with appropriate shelf life dates. Shelf life expiration dates of pre-mixed kits shall be based on the ingredient(s) with the shortest shelf life or the earliest expiration date whichever is earlier.

4.5.12 Quality controls for each molding compound batch for casting or assembling plugs to housings.

4.5.12.1 Molding compound batch definition. A batch of molding compound is herein defined as that quantity of compound mixed at one time using one combination of base resin and hardener lots from one manufacturer that have been tested together and met the requirements of MIL-M-48245 and which is used to assemble plugs to housings in one unchanged process. When a continuous type mixing process is used, a batch is defined as that quantity of molding compound mixed and dispensed in one continuous unchanged process during a four (4) hour period or portion thereof using one conbination of base resin and hardener lots fro one manufacturer that have been tested together and met the requirements of MIL-M-48245.

4.5.12.2 Resin/hardener wieght ratio control.

a. Hand mixing: For each batch of molding compound mixed at one time, the hardener and resin shall be weighed separately to verify that the ratio is between 1.8/1 and 2.2/1 before use.

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b. Automatic mixer method: At the beginning of production and once each four (4) hours during production two test shots shall be drawing with the mixer set to expel the hardener and resin separately. The hardener and resin shall be weighed to verify that the ratio is between 1.8/1 and 2.2/1. If the shots fail to meet the requirement, all wedges filled since the last acceptable test shall be rejected. Further, the mixer shall be corrected and shall pass the above test before being returned to use.

4.5.12.3 Cure contact. Two test bars from each batch of molding compound shall be molded at the same time the batch is used to assemble plugs to housings. The test bars shall be 0.5 inch diameter by 1.0 inch long or 0.5 inch square by 1.0 inch long. Test bar molding compound which the bars represent., i.e. the bars shall be cured in the same time periods as the items they represent. After completion of the cure, one bar shall be tested for ultimate compressive strength as specified in ASTM-D-695. The ends shall be flat and parallel to within .003 inch, and shall be perpendicular to the long axis of the test bar. Speed of testing shall be as specified in the ASTM. If the results of this test are below the limit specified in 3.9, the munitions represented by the bar shall be rejected. However, if the test bar failure can be attributed to a void, which is verified by the Government inspector, the second test bar (cast at the same time and representing the same batch of assemblies) shall be tested as a replacement.

4.5.13 Temperature control during molding/cure operations. Whenever the housing or munition is subjected to heating in ovens, temperture records shall be maintained and reviewed to assure that the minimum temperature has been reached and that the minimum temperture has not been exceeded (see 6.4). If any units have been subjected to temperatures beyond the maximum limit at any time, they shall be rejected. Temperature measuring instruments shall be approved and calibrated in accordance with 4.4.4.

4.5.14 Static functioning test of kill mechanism and overlay assembly. The sample kill mechanism and overlay assembly (KM&O) shall be assembled into an ADAM firing train housing, provided KM&O can be properly seated. The KM&O assembly shall be assembled into the housing, timing, fusing (H.T.F.) housing using normal assembly techniques. The booster cup of the detonating cord assembly shall be aligned with, and positioned against the output side of the lead cup assembly in the S&A. The second detonating cord assembly shall be placed in the same position as the first (see attached drawing), resting directly on top and the cord shall bend 90° upward, protruding vertically out of area to be closed and bonded. The booster keeper and the retaining clips shall be assembled. The

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H.T.F. housing shall then be closed, using normal assembly procedures. After curing, those H.T.F. assemblies to be tested at temperature extremes shall be temperature conditioned for two hours minimum at the temperature specified in Table II. The H.T.F. assemblies shall be tested in the orientations and the temperature specified in Table II. The testing procedures in the following paragraphs shall apply unless otherwise specified by the Contracting Officer. Test procedures and equipment designs shall be documented and submitted for approval (see 4.4.4 and 6.3).

In addition inspection shall be performed to assure that the safety rod is present and the piston actuator and detonator shorts have continuity. The kill mechanism and overlay assembly shall be assembled into the test housing using normal assembly techniques. The booster cup of the detonating cord assembly shall be aligned with, and positioned against the output side of the lead cup assembly in the OC S&A. The booster keeper and the retaining clips shall be assembled. The test housing shall then be closed using normal assembly procedures. After curing, those test wedges conditioned for two hours minimum at the temperature specified in Table 2. The test wedges shall be tested in the orientations and at the temperatures specified in Table 2. The testing procedures in the following paragraphs shall apply unless otherwise specified by the Contracting Officer. Test procedures and equipment designs shall be documented and submitted for approval (see 4.4.4 and 6.3).

4.5.14.1 Test arena. The arena or room shall be not less than twenty (20) feet square and fifteen (15) feet in height. It shall be contructed of material that will resist the blast of twenty-five (25) grams of high explosive. The walls and ceiling shall be free of areas where fragments, parts, etc., could lodge and be lost. At least one wall shall be constructed to permit obvservation of functioning. A camera shall be positioned with a viewing port set at a height above the test surface to photograph item functioning in front of a suitable grid for subsequent examination and evaluation. Alternate construction may be used with the approval of the contracting officer.

Provisions shall be made to determine the velocity of the kill mechanism by determining the distance it travels and measuring the time from wedge breakup to the audible report from kill mechanism functioning.

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4.5.14.2 Firing lines. A set of firing lines shall be provided and they shall be expendable if damaged. Undamaged lines may be reused for subsequent tests. The firing lines shall be color coded or suitably identified to preclude mixing. One set of lines shall be for the piston actuator and the other set shall be for the electric detonator. The lines shall be shielded within the test arena.

4.5.14.3 Firing equipment. The firing equipment shall:

a. Be capable of checking the continuity of the firing lines.

b. Provide for control of all camera stations and for starting and stopping an electronic digital counter.

c. Provide an output to the electric detonator that is approximately equivalent in magnitude and duration to that provided by the tactical wedge circuitry.

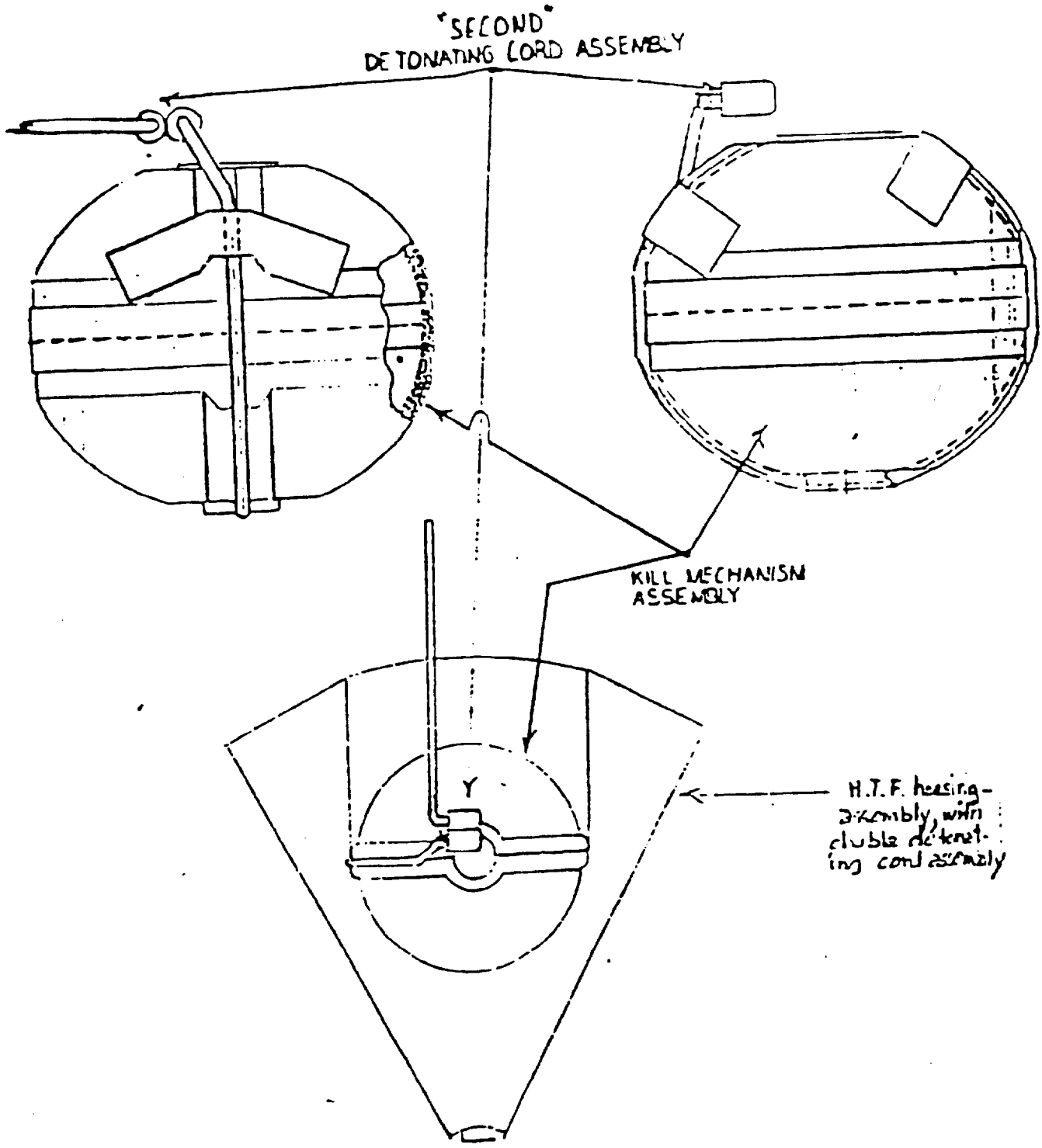
4.5.14.4 Test procedures. The H.T.F. assemblies shall be placed on a test pad of dry sand (sand capable of free flow at time of test). The firing line set shall be shorted. An M100 detonator shall be placed on the protruding detonating cord assembly, the firing line set will be attached to the detonator. The operator shall then exit from the test arena.

Following completion of all procedures for preparation of the H.T.F. assembly, functioning shall be accomplished using the following procedures.

The camera shall be positioned so that its field of view will record the entire functioning sequence of the H.T.F. assembly. The camera and firing line shall then be connected to the firing equipment. The electronic detonator shall then be connected to the firing equipment. The electronic detonator shall be fired. Any H.T.F. assembly which fails to function the KM&O assembly due to detonator failure, breakage of detonating cord assembly shall be considered a no test item and a replacement H.T.F. assembly shall be tested in its place. (See Figure 1).

4.5.14.5 Test evaluation (see 3.4). The photograph shall be examined to determine compliance with the requirements of 3.4. All residue from the test shall be collected. If any large fragments of the kill mechanism are found which exceed the dimensional value specified in 3.4 in any direction, the KM & O shall be considered to have failed the test. All test results and any other pertinent information gathered during the test shall be forwarded as specified by Data Item Description (DI-R-1724).

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MINE TEST LAYOUT

FIGURE 1

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a. Height of functioning of main charge (see 3.4c).

b. Fragmentation (compliance or non-compliance with 3.4b).

c. Average height of burst for the sample (see 3.4d).

d. A recored of the "no test" test housings including the failure mode, e.g. piston actuator, detonator, or test equipment.

e. A record of any unusual visual or audible conditions when the main charge functions. (e.g. yellow or white cloud indicating low order function vs black cloud indicating high order functioning).

f. Velocity of the kill mechanism assembly.

4.5.14.6 Maximum number of tests. The following criteria will be used to limit the number of times a delay detonator or propellant nut can be tested:

a. The combination of propellant nut lot and delay detonator lot accepted according to 4.5.5 shall be tested with the Kill Mechanism and Overlay.

b. The combination of propellant nut lot and delay detonator lot which passing the functioning test as outlined in 4.5.14 to 4.5.14.6 above shall be used to assemble the ADAM mine.

c. Each propellant nut lot and delay detonator lot which fails the functioning test as outlined in 4.5.14 to 4.5.14.6 above may be retested with two additional lots of delay detonators or propellant nut lots respectively provided criteria as outlined in 4.5.5.3(c) and 4.5.5.3(d) has not been exceeded.

d. Any propellant nut lot or delay detonator lot which fails the functioning tests as outlined in 4.5.14 to 4.5.14.6 two or more times will be rejected.

e. An additional test may be conducted using double sampling, (i.e., 100 samples; accept on 6, reject on 7) when failure occurs and investigation by the contractor, as verified by the Government Quality Assurance Representative, conclude the causes for failure were other than the propellant nut or delay detonator.

5. PACKAGING

Not applicable.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The assemblies covered by this specification are intended for use in the Projectiles, 155mm, M692/M731.

6.2 Acquisition requirements. See MIL-A-48078.

6.3 Submission of acceptance inspection equipment designs. (See MIL-A-48078), Equipment designs shall be submitted as required to: Commander, AMCCOM, ATTN: AMSMC-QAT-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Documentation of molding process controls. The contractor's documentation of molding process shall specify, as a minimum, controls and procedures for the folllowing:

a. Confirmation of accuracy of temperature-measuring instruments, including the method, location and frequency basis for obtaining temperature checks of all the temperature recording instruments for all the molding and curing phases.

b. Homogeneity of molded munitions, to preclude mixing of munitions from different lots during molding and cure.

c. The storage of permanent records, available for Government review, of the time-temperature cycle for all phases of the molding and cure process and the items in a and b above. The records shall provide suitable identification by molding compound batch and lot number as appliable.

6.5 Bulk density of explosive composition A5. The purpose of maintaining the bulk density of the explosive composition A5 at 0.83 to 1.00 is to remove any excessively fine or coarse particles when present in certain batches. Excessive concentrations can cause low order function.

The bulk density requirements can be met by screening the powder using a Standard US Sieve #20 and over #50 US Sieve. Only the powder particles that are retained on the #50 sieve are used. The powder that falls through the #50 sieve in considered too fine and is scrapped. The powder retained on the #20 sieve is considered too course and is also scrapped.

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It is also recommended that the particle size distribution of each batch of powder to be determined before and after the above sieving using a 100 gram sample of powder and using US laboratory sieves #20, 30, 40, 50, 60, 70, 80, 100, 140 and 200 and that the data obtained be included with the bulk density certification.

6.6 Safety precautions for molding compound and sealing compound. When handling, mixing or curing the molding or sealing compound or its constitutents, suitable precautions must be taken, as some of the constitutents and compounds are toxic and exude toxic fumes when mixed or heated. Use adequate ventilation. Avoid contact with skin, eyes and clothing and prolonged or repeated breathing of fumes.

6.7 Toxicity of liquid propellent. See MIL-P-48333.

6.8 Data cards. In addition to the normal distribution of records, one copy of the ammunition data card for each munition lot shall be forwarded to: Commander, AMCCOM, ATTN: AMSMC-QAT-M (D), Picatinny Arsenal, NJ 07806-5000.

6.9 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.10 Data requirements. Deliverable data required by this specification is cited in the paragraphs listed below. The form DD1423 should include distribution to AMSMC-QAT-M (D).

Paragraph	Data Requirement	Applicable DID
3.3	Plan for controls in use of Adhesives and Sealants	DI-T-5372 (Tailored)
3.6	Plan for Safety in Handling and Processing of Munitions	DI-T-5372 (Tailored)
3.8	Plan for Molding and Cure Process Controls	DI-T-5372 (Tailored)
4.4.2	Quality Inspection Defect Report	DI-R-1721
4.5.14.6	Quality Inspection Test Demonstration and Evaluation Report	DI-R-1724
6.8	Ammunition Data Card	DI-MISC-80043

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6.11 Subject term (keyword) listing.

Arena Tests.
Burst Height.
Delay Detonator.
Kill Mechanisms.
Mine, Anti-Personnel.
Molding Compound.
Propellant Nut.

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian
Army-AR

Preparing activity:
Army-AR

(Project 1320-AB15)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-48186B (AR)	14 November 1990

3. DOCUMENT TITLE
MUNITION FOR PROJECTILE, 155MM, HE, M692/M731, LOADING AND ASSEMBLING OF

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US Army ARDEC Standardization & Specification Office	(1) Commercial (201) 724-6671	(2) AUTOVON 880-6671
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 — Previous editions are obsolete. — 198/290

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